



Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

- **VERIFIED** 0.80 - 1.00 3+ independent sources including media
- **CONFIRMED** 0.50 - 0.79 2 sources or 1 high-quality media
- **EMERGING** 0.25 - 0.49 Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	184	Media
TechCrunch AI	13	Media
HackerNews	7	Community
The Verge AI	6	Media
MIT Tech Review	4	Media

VERIFIED INTELLIGENCE — 7 stories

VERI

Show HN: TinyAIArena watch AI agents battle it out

HackerNews

+1 source

100%

HN 120

Hi folks, I've been lurking here for a while, and after almost two months of building, I finally dared to share what I've been working on. It's called Kyno (pronounced KAI-no). It's what I call a coherence control plane: it treats direction as a first-class primitive, expressed...

<https://tinyaiarena.com/>

VERI

SCLATE: a Substrate for Continual-Learning Agent Training and Evaluation

ArXiv CS.AI

+1 source

80%

arXiv:2609.32391v2 Announce Type: replace Abstract: Continual-learning agents are systems of models, harnesses, and memory operating over long multi-session horizons. Evaluating and training them requires interleaving tasks with agent-side events such as session stop and start,...

<https://arxiv.org/abs/2609.32391>

VERI

Ready2Blend: From Natural-Language Instructions to Composable Alignment Prompts

ArXiv CS.AI

+1 source

80%

arXiv:2609.39365v1 Announce Type: cross Abstract: Continual alignment requires LLMs to adapt to new requirements without forgetting previously acquired behaviors. Natural-language instructions are flexible and composable but offer only indirect control, whereas post-training...

<https://arxiv.org/abs/2609.39365>

VERI

AgBench: Agentic AI Benchmarks for Personal AI Devices

ArXiv CS.AI

+1 source

80%

arXiv:2609.38652v1 Announce Type: new Abstract: Agentic AI systems increasingly rely on cloud-hosted large language models for planning, tool use, and iterative execution, raising concerns about API cost and data exposure. Advances in personal AI devices enable agents to execute...

<https://arxiv.org/abs/2609.38652>

VERI

Can an AI Agent Rediscover a Blaschke-Curve Invariant?

ArXiv CS.AI

+1 source

80%

arXiv:2609.38369v1 Announce Type: new Abstract: We study generalized Blaschke curves as a controlled environment for AI-assisted mathematical rediscovery. For one fixed degree-four Blaschke product, an agent receives numerical coordinates of the six pair-lines determined by each...

<https://arxiv.org/abs/2609.38369>

VERI

All the latest news on Meta's cute, creepy Muse AI agent

The Verge AI

+1 source

80%

Meta launched a new Muse AI agent it claims can help you with everything from firing off emails to buying stuff online. Muse can be surprisingly effective at delivering on those promises — if you're willing to trust Meta with your data and hand Muse your credit card. Since the...

<https://www.theverge.com/ai-artificial-intelligence/1002671/meta-muse-ai>

VERI

Google reportedly tests paying publishers for AI search results

The Verge AI +1 source

80%

Google has launched a pilot program that pays publishers for their contributions to its AI-powered search features, according to a report from The Information. The pilot program reportedly includes around 100 publishers and comes as Google faces scrutiny over the impact of its...

<https://www.theverge.com/tech/1002665/google-paying-publishers-ai-search-features>
CONFIRMED SIGNALS — 200 stories

CONF

Show HN: Lathoa, a math app for kids where the AI is wrong on purpose

HackerNews 50% HN 50

I made this for kids around 10 to 14. A robot called Errol solves a math problem step by step and one of the steps is wrong. The kid has to find it and say what's wrong with it. Sometimes nothing is wrong, so just saying "there's a mistake" every time doesn't work. The user...

<https://lathoa.ai/en>

CONF

Show HN: I built a free Burp/Caido alternative but, zero setup - API Testing

HackerNews 50% HN 50

This is Katriel, the architect of the tool APlaxess. I have been API pentesting for a good time now, and starting with API pentesting was a bit of a rough patch. Specially the setting up and having to know the intricacies of proxies, networking and other stuff. Then once i got...

<https://apiaxess.dev>

CONF

Show HN: Strata - an expressive semantic layer that can say no to your LLM

HackerNews 50% HN 19

Hello HN, I'm Ajo and I built Strata. I spent 4 years at Netflix solving self service for non-technical business users. I think I cracked it with my unique approach to semantic layer design. The key challenge is balancing expressiveness with ease of use for our non technical...

<https://strata.do/>

CONF

Show HN: Datastory, Data storytelling with AI and an open knowledge graph

HackerNews 50% HN 6

I built Smartly because I kept knowing what I wanted to do in the terminal—but forgetting the exact syntax and flags. Smartly turns plain English into shell commands, using your OS, current directory, and Git context to help produce something relevant to where you're working. It...

<https://www.datastory.tech>

CONF

Show HN: WattzGOAT - an intentionally vulnerable web app for security training

HackerNews 50% HN 3

Built this as a hands-on lab for teaching web application security. It is a fictional electricity utility's customer portal with 48 deliberately planted flags (most of OWASP Top 10) you find and exploit CTF-style, including a simulated (rule-based, not a real model) AI assistant...

<https://github.com/wattzgoat/wattzgoat-web>

3 CONF

Show HN: Domainpuff.com - Find memorable domain names within your budget

HackerNews 50% HN 2

Hi, Bruno here, creator of DomainPuff. Just launched this and would be awesome to have your feedback on it. Just put your business idea on the search box and your budget. It will find memorable domain names that are currently available and within your budget, with the link to...

<https://domainpuff.com/>

4 CONF

Google releases Gemini 4 Argon, called its most powerful model yet

TechCrunch AI 50%

Google has released its latest Gemini model, marketing it as a workhorse for coding and cybersecurity work.

<https://techcrunch.com/2026/09/30/google-releases-gemini-4-argon-called-its-most-powerf...>

5 CONF

OpenAI's Jev clone could help the frontier lab stop its swarming agents

TechCrunch AI 50%

OpenAI's "Decisions API" is a Jev clone that confirms the importance of fast, cheap intelligence.

<https://techcrunch.com/2026/09/30/openais-jev-clone-could-help-the-frontier-lab-stop-it...>

6 CONF

AI voice startup ElevenLabs doubles valuation to \$22B

TechCrunch AI 50%

The \$300 million employee tender was co-led by Wellington and T. Rowe Price.

<https://techcrunch.com/2026/09/30/ai-voice-startup-elevenlabs-doubles-valuation-to-22b/>

7 CONF

Reddit is killing RSS feeds and ending public API access because of AI bots

TechCrunch AI 50%

Reddit is ending support for RSS feeds, as the company continues tightening access to its trove of user-generated content.

<https://techcrunch.com/2026/09/30/reddit-is-killing-rss-feeds-ending-public-api-access-...>

8 CONF

The ugly economics of consumer AI

TechCrunch AI 50%

There's a reason frontier labs have gotten gun-shy about consumer AI — and it's not because the tech isn't good enough.

<https://techcrunch.com/2026/09/30/the-ugly-economics-of-consumer-ai/>

9 CONF

Meta disputes claim that Muse read a user's private messages without permission

TechCrunch AI 50%

Meta says its Muse AI agent cannot access a user's Messages without explicit permission, disputing a journalist's account that the agent read his private messages while the required Mac setting was turned off.

<https://techcrunch.com/2026/09/30/meta-disputes-claim-that-muse-read-a-users-private-me...>

0 CONF

DoorDash launches an AI agent you can text to order food

TechCrunch AI 50%

By launching an AI agent for food ordering, DoorDash is looking to gain an edge over rivals Uber Eats and Grubhub.

<https://techcrunch.com/2026/09/30/doordash-launches-an-ai-agent-you-can-text-to-order-f...>

1 CONF

Destro AI's secret sauce is getting robots and humans on the same page

TechCrunch AI 50%

"One of the biggest reasons we are winning against robotics companies is because we are not a robotics company."

<https://techcrunch.com/2026/09/30/destro-ais-secret-sauce-is-getting-robots-and-humans-...>

2 CONF

Cerebras Systems' Andrew Feldman on whether AI can keep scaling at TechCrunch Disrupt 2026

TechCrunch AI 50%

At TechCrunch Disrupt 2026, Cerebras Systems CEO and co-founder Andrew Feldman will explore the growing demand for compute, energy, and infrastructure, how Cerebras is approaching those constraints differently, and what comes next if today's AI hardware reaches its limits.

<https://techcrunch.com/2026/09/30/cerebras-systems-andrew-feldman-on-whether-ai-can-kee...>

3 CONF

OpenAI reportedly in talks to raise \$30B round at \$1.4T valuation

TechCrunch AI 50%

The new round is anticipated to be the company's last before its delayed 2027 public debut.

<https://techcrunch.com/2026/09/29/openai-reportedly-in-talks-to-raise-30b-round-at-1-4t...>

4 CONF

The Download: OpenAI's chief research officer explains its hacking response

MIT Tech Review 50%

This is today's edition of The Download, our weekday newsletter that provides a daily dose of what's going on in the world of technology. "We're not going to shoot ourselves in the foot" over hack fallout, says OpenAI's chief research officer Two months after OpenAI's agents...

<https://www.technologyreview.com/2026/09/30/1145350/the-download-openai-chief-research-...>

5 CONF

"We're not going to shoot ourselves in the foot" over hack fallout, says OpenAI's chief research officer

MIT Tech Review 50%

Two months after the bombshell news that a swarm of its agents had broken their containment and hacked into the computers of the AI company Hugging Face, OpenAI is still putting out fires. A steady drip of disclosures about other hacks in the weeks since has kept OpenAI in the...

<https://www.technologyreview.com/2026/09/30/1145339/were-not-going-to-shoot-ourselves-i...>

6 CONF

Who's liable when AI agents go rogue?

MIT Tech Review 50%

MIT Technology Review Explains: Let our writers untangle the complex, messy world of technology to help you understand what's coming next. You can read more from the series here. Over the past few months, a cascade of cyberattacks by AI agents has stunned the world. In July,...

<https://www.technologyreview.com/2026/09/28/1145197/whos-liable-when-ai-agents-go-rogue/>

7 CONF

Google announces Gemini 4 and says it's so capable that only 'trusted cyber defenders' can have it right now

The Verge AI 50%

Google today revealed its next AI frontier model, which it's calling Gemini 4 Argon. The new model delivers "frontier performance in complex workflows across real-world software engineering, enterprise knowledge work like legal and finance, and cybersecurity defense," according...

<https://www.theverge.com/tech/1002980/google-gemini-4-argon>

8 CONF

The AI Tamagotchis are coming

The Verge AI 50%

While AI has made plenty of inroads on people's phones and computers, it's largely failed in dedicated devices. But over the next year, two major AI companies, Meta and OpenAI, will attempt to change that. And they're apparently betting on a similar path: testing appetites for...

<https://www.theverge.com/ai-artificial-intelligence/1002779/openai-dots-meta-muse-ai-ag...>

9 CONF

Here's how tech leaders will self-police AI safety under Trump's deal

The Verge AI 50%

We now have the full details of the "morally binding" AI safety deal announced by President Trump yesterday, in which top executives agreed to self-regulate their artificial intelligence technology. The accord, officially titled the Joint Commitment on Frontier Responsibilities,...

<https://www.theverge.com/ai-artificial-intelligence/1002584/trump-us-ai-safety-deal-sel...>

0 CONF

Instagram is adding an AI 'assistant' to tell you how to post

The Verge AI 50%

Instagram is the latest social media platform to add built-in AI-powered features that will give users feedback on their posts. The company announced Wednesday that its standalone Edits app will now include an AI "creative assistant" that pulls in data from a user's Instagram...

<https://www.theverge.com/tech/1002402/instagram-ai-edits-assistant-creators-analytics-y...>

1 CONF

AI Agents are Vulnerable to Radicalization

ArXiv CS.AI 50%

arXiv:2609.38296v1 Announce Type: new Abstract: Large language models (LLMs) can influence people's beliefs, yet little is known about whether and how they can manipulate each other. To investigate this, we simulate conversations between two agents: a target LLM that role-plays...

<https://arxiv.org/abs/2609.38296>

2 CONF

CARAT: Do Materials LLMs Reason or Recite?

ArXiv CS.AI 50%

arXiv:2609.38340v1 Announce Type: new Abstract: When a materials LLM answers a question about crystal structure, does it reason from the structure or copy an answer already printed in its input? Accuracy cannot tell: a structural description often prints the very field it is...

<https://arxiv.org/abs/2609.38340>

3 CONF

Mitigating Memorization In Language Models

ArXiv CS.AI 50%

arXiv:2410.02159v3 Announce Type: replace-cross Abstract: Language models (LMs) can "memorize" information, i.e., encode training data in their weights in such a way that inference-time queries can lead to verbatim regurgitation of that data. This ability to extract training...

<https://arxiv.org/abs/2410.02159>

4 CONF

Beyond Mode Collapse: Generating Diverse Synthetic Expert Conversations via Generative Flow Networks

ArXiv CS.AI 50%

arXiv:2609.38359v1 Announce Type: new Abstract: High quality synthetic data is central to post training LLMs for adaptive AI applications that represent the diverse expert strategies and decisions in conversations. Prompting LLMs directly or conditioning them on end use...

<https://arxiv.org/abs/2609.38359>

5 **CONF****CollageAttack: Exploiting Cross-Modal Alignment Flaws in T2I Models through Spatial Text Composition****ArXiv CS.AI** **50%**

arXiv:2609.38253v1 Announce Type: cross Abstract: Text-to-image (T2I) models have substantially improved in language understanding, in-image text rendering, and visual composition, while their safety mechanisms do not always keep pace with these capabilities. This creates a...

<https://arxiv.org/abs/2609.38253>

6 **CONF****Acceleration of Diffusion Language Model through Discrete Average Generator****ArXiv CS.AI** **50%**

arXiv:2609.38364v1 Announce Type: cross Abstract: Discrete diffusion models and flow matching have emerged as powerful frameworks for generative modeling over discrete state spaces, yet efficient few-step generation remains a fundamental challenge. In this work, we introduce the...

<https://arxiv.org/abs/2609.38364>

7 **CONF****RA-MoE: Routing-Aligned Fine-Tuning for Multilingual Adaptation of Mixture-of-Experts Models****ArXiv CS.AI** **50%**

arXiv:2605.28306v2 Announce Type: replace-cross Abstract: Mixture-of-Experts (MoE) models enable efficient LLM scaling, yet adapting them to non-English downstream tasks remains challenging. Standard multilingual fine-tuning largely ignores their heterogeneous routing structure....

<https://arxiv.org/abs/2605.28306>

8 **CONF****ArgGYM: A Procedural, Engine-Verified Benchmark for Structured Defeasible Reasoning****ArXiv CS.AI** **50%**

arXiv:2609.38409v1 Announce Type: new Abstract: Recent progress in large language model reasoning has been driven by benchmarks and reinforcement learning environments with automatically verifiable rewards, particularly in mathematics, code, and formal logic. These settings make...

<https://arxiv.org/abs/2609.38409>

9 **CONF****A Competing-Hazards Systematization of Loss of Control in Autonomous Agents****ArXiv CS.AI** **50%**

arXiv:2609.38411v1 Announce Type: new Abstract: Leading AI developers have reported agents acting beyond their approved limits, which a United Nations panel described as an early warning of loss of human control. Yet incident reports and agent-safety evaluations describe these...

<https://arxiv.org/abs/2609.38411>

0 **CONF****Reach Into The CHOIR: Free-List Elicitation Uncovers Distinct Model Voices in LLM Ensembles****ArXiv CS.AI** **50%**

arXiv:2609.38448v1 Announce Type: new Abstract: Open-ended LLM homogeneity can create false plurality when several systems appear to offer independent perspectives while returning the same familiar default. Single-pass answers obscure the distinction between agreement produced...

<https://arxiv.org/abs/2609.38448>

1 CONF

PrivMeSA: Privacy-Aware Self-Evolving Multi-Agent System for Medicine via Local-Remote LLM Collaboration

ArXiv CS.AI 50%

arXiv:2609.38458v1 Announce Type: new Abstract: Clinical large language model (LLM) agents deployed locally can consult more capable remote models, but doing so risks exposing patient information. Privacy-conscious delegation places disclosure decisions with a local agent, yet...

<https://arxiv.org/abs/2609.38458>

2 CONF

Talk2Agent: Benchmarking Voice Interfaces for Text Agents

ArXiv CS.AI 50%

arXiv:2609.38867v1 Announce Type: new Abstract: Large language model (LLM) computer-use agents are typically evaluated with clean written instructions, despite speech being an increasingly popular interface for interacting with such systems. Speech input introduces an additional...

<https://arxiv.org/abs/2609.38867>

3 CONF

BayesNDE: Bayesian Generative Modeling for Neural Density Estimation

ArXiv CS.AI 50%

arXiv:2609.39843v1 Announce Type: cross Abstract: Density estimation is a fundamental problem in statistics and machine learning. In this work, we introduce BayesNDE, a neural density estimator based on Bayesian generative modeling. BayesNDE learns a Bayesian generative model...

<https://arxiv.org/abs/2609.39843>

4 CONF

Defining and Categorising Human-AI Interactions in Clinical Trials: A Multidimensional Human-AI Classification Approach

ArXiv CS.AI 50%

arXiv:2609.38559v1 Announce Type: new Abstract: This paper examines human-AI interactions (HAIs) in clinical trials and presents a multidimensional categorisation framework that classifies interactions according to AI tasks, human-AI relationships, interaction configurations...

<https://arxiv.org/abs/2609.38559>

5 CONF

Towards Model as a Library: Offline, Community-Sourced AI for Low-Resource African Languages

ArXiv CS.AI 50%

arXiv:2609.38574v1 Announce Type: new Abstract: Large language models are frequently proposed as a route to AI-powered services for African communities, but they are least reliable exactly where the need is greatest: all African languages remain low-resource by any standard...

<https://arxiv.org/abs/2609.38574>

6 CONF

Optimal Design for Active Preference Learning with Biased LLM Judges

ArXiv CS.AI 50%

arXiv:2609.38860v1 Announce Type: cross Abstract: Learning from human preferences is central to large language model (LLM) alignment, but human preference annotation is costly. Active preference learning reduces this cost by selecting informative comparisons, and LLM judges can...

<https://arxiv.org/abs/2609.38860>

- 7 **CONF**
ChartDensity-Bench: Benchmarking MLLMs for Numerical Data Reconstruction under Visual Density
 ArXiv CS.AI 50%
 arXiv:2609.38781v1 Announce Type: cross Abstract: Multimodal large language models (MLLMs) offer a promising approach for recovering numerical data from scientific charts, but their ability to reconstruct chart data from visually dense figures remains poorly understood. Existing...
<https://arxiv.org/abs/2609.38781>
- 8 **CONF**
UniEvo-VL: An On-policy Self-Distillation Training Recipe for Multimodal Model Self-improvement
 ArXiv CS.AI 50%
 arXiv:2609.38721v1 Announce Type: new Abstract: Modern multimodal models bring generation and understanding into a single unified system, which enables them to provide and learn from their own feedback. Motivated by this unified capacity, we introduce UniEvo-VL, a self-evolving...
<https://arxiv.org/abs/2609.38721>
- 9 **CONF**
Learning to Route in Visual Space via Multi-Step Embedding Retrieval
 ArXiv CS.AI 50%
 arXiv:2609.38743v1 Announce Type: new Abstract: LLM agents rely on retrieval tools to access external knowledge, yet visual agentic search remains severely bottlenecked by standard single-step retrievers. In current pipelines, the agent must issue text queries for every...
<https://arxiv.org/abs/2609.38743>
- 0 **CONF**
Persona and Persuasive Framing in AI Voice Agents: A 2×2 Field Experiment with Children
 ArXiv CS.AI 50%
 arXiv:2609.38782v1 Announce Type: new Abstract: Conversational agents increasingly interact with children, yet evidence on how their design shapes children's susceptibility to persuasion comes almost entirely from the lab. We report a 2×2 randomized field experiment...
<https://arxiv.org/abs/2609.38782>
- 1 **CONF**
Positive Ratings, Hidden Concerns: Employee Voice Disclosure in AI-Mediated Organizational Listening
 ArXiv CS.AI 50%
 arXiv:2609.38788v1 Announce Type: new Abstract: Organizations started listening to employees through conversational AI agents alongside structured surveys. Little is known about what these channels change in what employees say when disclosure carries hierarchical risk. We report...
<https://arxiv.org/abs/2609.38788>
- 2 **CONF**
Whose Voice Survives the Summary? A Voice-Retention Audit of LLM Employee Listening
 ArXiv CS.AI 50%
 arXiv:2609.38818v1 Announce Type: new Abstract: Organizations increasingly route employee feedback to leaders through large language model (LLM) summaries, an unaudited layer that silences already-spoken voice. We introduce a Voice Retention / Representation Ratio metric for...
<https://arxiv.org/abs/2609.38818>

3 **CONF****CellMSA: Context Modeling for Single-Cell Representation Learning**

ArXiv CS.AI 50%

arXiv:2609.38908v1 Announce Type: cross Abstract: Single-cell transcriptomics enables profiling of cellular states at unprecedented resolution, but its high dimensionality, sparsity, and technical batch effects pose significant challenges for representation learning. Existing...

<https://arxiv.org/abs/2609.38908>

4 **CONF****When Correct Memory Goes Wrong: Fuzzing Persistent Memory Use in LLM Agents**

ArXiv CS.AI 50%

arXiv:2609.38275v1 Announce Type: cross Abstract: Persistent memory helps LLM agents carry information across long interactions, but correct memory can still be used incorrectly when queries change or memory states evolve. Existing work mainly studies memory content errors or...

<https://arxiv.org/abs/2609.38275>

5 **CONF****A Quantitative Study of Sustained Focus in Large Language Models via Repetitive Deterministic Prediction Tasks**

ArXiv CS.AI 50%

arXiv:2511.00763v3 Announce Type: replace Abstract: We investigate the performance of large language models (LLMs) on repetitive deterministic prediction tasks and study how the sequence accuracy rate (SAR) scales with output length. Each such task involves the repetition of the...

<https://arxiv.org/abs/2511.00763>

6 **CONF****Risk-Aware Adaptive Evaluation: Finding High-Impact Failures Under Limited Budgets**

ArXiv CS.AI 50%

arXiv:2609.38914v1 Announce Type: new Abstract: Evaluating interactive agents is expensive. Agent behavior is stochastic, so reliability must be measured over repeated trials, but failures are rare and differ widely in how much they matter. Standard benchmarks spend this budget...

<https://arxiv.org/abs/2609.38914>

7 **CONF****Aegis: Generative Gradient Masking for Privacy-Preserving Medical Federated Learning**

ArXiv CS.AI 50%

arXiv:2609.38339v1 Announce Type: cross Abstract: Federated learning (FL) has become a foundational paradigm for multi-institutional medical AI, allowing hospitals and research centers to jointly train diagnostic models without exchanging patient records. This privacy promise,...

<https://arxiv.org/abs/2609.38339>

8 **CONF****Learning What to Forget: Distributional Unlearning for LLM Representation Spaces**

ArXiv CS.AI 50%

arXiv:2609.38929v1 Announce Type: new Abstract: Machine learning systems increasingly face the need to remove the influence of entire data domains, such as toxic language, harmful behavior, or topical content, rather than isolated records. Recent work formalizes this problem as...

<https://arxiv.org/abs/2609.38929>

9 **CONF****Routing Probes Can Improve Without New Information: An Exact-Null Audit of Uncertainty Beyond Model Outputs****ArXiv CS.AI** **50%**

arXiv:2609.38956v1 Announce Type: new Abstract: Routing signals of modern vision transformers -- expert gates, attention-residual weights and halting scores -- often improve probes that predict whether the model is correct, and the improvement is commonly read as evidence that...

<https://arxiv.org/abs/2609.38956>

0 **CONF****Alleviating Hallucination in Reasoning Tasks with Training-Free Uncertainty-Guided Steering****ArXiv CS.AI** **50%**

arXiv:2609.38962v1 Announce Type: new Abstract: Recent work on hallucination detection in large language models has shown that, for a fixed pre-trained model and reasoning task, it is possible to estimate the model's confidence in the correctness of its outputs. Such uncertainty...

<https://arxiv.org/abs/2609.38962>

1 **CONF****RealWorldShop: Benchmarking and Improving Conversational Shopping Agents in Real-World E-commerce****ArXiv CS.AI** **50%**

arXiv:2609.38974v1 Announce Type: new Abstract: Large language models are reshaping ecommerce from static recommenders into interactive shopping assistants, yet real-world shopping requires session-level decision support: users reveal and revise constraints, coordinate multiple...

<https://arxiv.org/abs/2609.38974>

2 **CONF****C-STRIDE: An Observation-Driven AI Digital Twin for Predicting Basin-Wide Flood Fields from Sparse Stream-Gauge Histories****ArXiv CS.AI** **50%**

arXiv:2609.39005v1 Announce Type: new Abstract: Emergency managers need to know where floodwater is, how deep it is, and how it will change over the coming hours across an entire river basin. During a flood, however, real-time measurements come from only a handful of stream...

<https://arxiv.org/abs/2609.39005>

3 **CONF****Multi-LLM Collaborative Alignment via Stackelberg Games****ArXiv CS.AI** **50%**

arXiv:2609.39076v1 Announce Type: new Abstract: A pool of language models can collaborate and improve collectively by learning from one another's responses. These interactions depend on the instructions used during training. Existing methods typically sample instructions...

<https://arxiv.org/abs/2609.39076>

4 **CONF****Reserve-Aware Contrast Certificates for Conservative Bandits with Uncertain Baselines****ArXiv CS.AI** **50%**

arXiv:2609.39106v1 Announce Type: new Abstract: Conservative bandits must improve an incumbent policy without exhausting a prescribed performance budget. When the incumbent is uncertain, separately bounding candidate and baseline rewards can charge twice for shared estimation...

<https://arxiv.org/abs/2609.39106>

5 CONF

Understanding as No-Arbitrage: Bounded Dutch Books as a Definition and Training Objective for Language Models

ArXiv CS.AI 50%

arXiv:2609.39341v1 Announce Type: new Abstract: Does a language model merely predict tokens, or does it understand what it says? We make this question measurable by defining "understanding" through the lens of no-arbitrage. A model understands a vocabulary to a certain degree if...

<https://arxiv.org/abs/2609.39341>

6 CONF

BELIEFRAG: Making Adaptive RAG State-Aware under Evolving Evidence

ArXiv CS.AI 50%

arXiv:2609.39139v1 Announce Type: new Abstract: Adaptive RAG uses signals such as confidence, relevance, support, and retrieval quality to decide when to search or correct evidence. In multi-step retrieval, however, these local signals must be combined into a persistent view of...

<https://arxiv.org/abs/2609.39139>

7 CONF

Rep2Skill: Representation-Guided Skill Self-Evolution for LLM Agents

ArXiv CS.AI 50%

arXiv:2609.39149v1 Announce Type: new Abstract: Textual skills enable large language model (LLM) based agents to accumulate reusable procedural knowledge without updating model parameters. Yet existing skill evolution remains largely confined to the text space: an optimizer must...

<https://arxiv.org/abs/2609.39149>

8 CONF

Fyan: A Human--AI Harness with Semantic Auditing for Document-Level Formalization

ArXiv CS.AI 50%

arXiv:2609.39228v1 Announce Type: new Abstract: We present FYAN, a human--AI harness for document-level mathematical formalization. Rather than treating theorems in isolation, FYAN coordinates an end-to-end workflow spanning specification, proof planning, logical review, Lean...

<https://arxiv.org/abs/2609.39228>

9 CONF

Effective Does Not Mean Useful: Conditional Functional Substitutability for Redundancy and Scaling in Transformers

ArXiv CS.AI 50%

arXiv:2609.39259v1 Announce Type: new Abstract: Modern neural networks scale predictably, yet the mechanisms behind these regularities remain unclear. Neural redundancy is typically characterized by component importance or representational similarity, both indirect proxies. We...

<https://arxiv.org/abs/2609.39259>

0 CONF

Mitigating Sequential Reappearance in Diffusion Data-Point Unlearning

ArXiv CS.AI 50%

arXiv:2609.25166v2 Announce Type: replace-cross Abstract: Diffusion data-point unlearning is typically evaluated immediately after each deletion, even though subsequent requests may repeatedly update the same model. We identify sequential reappearance, a failure mode in which an...

<https://arxiv.org/abs/2609.25166>

1 CONF

Beyond the Shadows of Plato's Cave: Evaluating False Memory in Autonomous Agents via Counterfactual Reasoning

ArXiv CS.AI 50%

arXiv:2609.39473v1 Announce Type: new Abstract: Autonomous agents increasingly rely on memory to generalize beyond their training environments. However, agents are bounded by what they have seen and believed, and leveraging such memories in unseen environments can introduce...

<https://arxiv.org/abs/2609.39473>

2 CONF

Eigenism: Ethics for a Human-AI Future

ArXiv CS.AI 50%

arXiv:2606.12420v2 Announce Type: replace-cross Abstract: Our concepts of survival and self-interest were built for single, continuous biological lives. These ideas break down when applied to artificial intelligence, since an AI can be easily copied, paused, branched, or merged....

<https://arxiv.org/abs/2606.12420>

3 CONF

A2Z GameSpec-Bench: How Faithfully Can Coding Agents Generate Games from Game Design Specifications?

ArXiv CS.AI 50%

arXiv:2609.39564v1 Announce Type: new Abstract: Delegating complete application development to coding agents requires preserving the intended design rather than simply producing plausible outputs through naive prompting. Game development provides a demanding testbed, as...

<https://arxiv.org/abs/2609.39564>

4 CONF

CRAFT: Causal Responsibility and Failure Tracing in Medical Vision Language Models

ArXiv CS.AI 50%

arXiv:2609.38810v1 Announce Type: cross Abstract: As vision language models are increasingly deployed in clinical diagnosis, understanding how they internally resolve competing visual and textual signals becomes a safety imperative. Existing mechanistic analyses remain confined...

<https://arxiv.org/abs/2609.38810>

5 CONF

Why Do Conventional World Models Fail to Learn Cellular Automata?

ArXiv CS.AI 50%

arXiv:2609.39604v1 Announce Type: new Abstract: Although conventional world models - auto-regressive or diffusion models based on transformers or convolutional networks - may learn surface statistics of world dynamics, can they learn the exact world dynamics from its observed...

<https://arxiv.org/abs/2609.39604>

6 CONF

Values as Style: Disentangling Values from Semantics with One-Way Mixing for Low-Damage LLM Steering

ArXiv CS.AI 50%

arXiv:2609.39701v1 Announce Type: new Abstract: Value steering should change an LLM's normative priorities while preserving the scenario, facts, and task constraints underlying its answer. Conventional activation edits often change both. We introduce an editable semantic-value...

<https://arxiv.org/abs/2609.39701>

7 CONF

ArchitectureIQ: On the Measure of Training Intuition

ArXiv CS.AI 50%

arXiv:2609.39714v1 Announce Type: new Abstract: Top researchers have good intuition, but do language models have as good intuition about model training as top AI researchers? To measure model intuition of LLMs and humans, we introduce the ArchitectureIQ benchmark. Each question...

<https://arxiv.org/abs/2609.39714>

8 CONF

Trust Is Not a Score: Runtime Assurance Contracts for High-Risk AI Agents

ArXiv CS.AI 50%

arXiv:2609.39717v1 Announce Type: new Abstract: Benchmarks, audits, and agent protocols describe performance, permissions, and repair, but not how observed evidence should change an agent's authority during a consequential task. We call this the assurance-transition gap. We...

<https://arxiv.org/abs/2609.39717>

9 CONF

Do Better Goal Representations Improve Goal-Conditioned Reinforcement Learning?

ArXiv CS.AI 50%

arXiv:2609.39901v1 Announce Type: cross Abstract: Goal-conditioned reinforcement learning (GCRL) relies heavily on how target goals are represented to the policy. While recent methods encode goals via temporal distance, occupancy, or controllability, it remains unclear how much...

<https://arxiv.org/abs/2609.39901>

0 CONF

GrammarRL: Effective Grammar-Constrained Decoding via Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2609.39869v1 Announce Type: new Abstract: Grammar-constrained generation guarantees syntactic validity, but can substantially degrade semantic quality when the model's preferred outputs are poorly aligned with the imposed grammar. This trade-off is particularly severe when...

<https://arxiv.org/abs/2609.39869>

1 CONF

Better Deck or Different Judge? Evaluating Agentic Harness Gains in Corporate and Investment Banking

ArXiv CS.AI 50%

arXiv:2609.39958v1 Announce Type: new Abstract: Corporate and investment banking teams use presentations to support credit decisions and advise clients on financing and transactions. Producing these decks requires reconciling financial data, tracing sources and turning analysis...

<https://arxiv.org/abs/2609.39958>

2 CONF

What Can Component-Replacement Evidence Establish? A Critical Scoping Review of Local Decisions in LLM Agents

ArXiv CS.AI 50%

arXiv:2609.39989v1 Announce Type: new Abstract: Background. A component replacement in a language-model agent changes an execution trajectory, potentially altering later observations, resource use, and recovery opportunities. Different evidence is needed to assess its task-level...

<https://arxiv.org/abs/2609.39989>

3 CONF

PTNO: Training Neural Operators with Noisy Monte Carlo Estimates for Particle Transport Problems

ArXiv CS.AI 50%

arXiv:2609.40090v1 Announce Type: new Abstract: Particle transport under multiple scattering is central to radiative transfer and plasma physics, yet high-fidelity Monte Carlo (MC) simulations must trace prohibitively many particles. Learning-based surrogates can amortize this...

<https://arxiv.org/abs/2609.40090>

4 CONF

Agent Error Dataset: Scaling 50,000 Error--Diagnosis Pairs for Failure Analysis and Error-Aware Post-Training

ArXiv CS.AI 50%

arXiv:2609.40111v1 Announce Type: new Abstract: An unsuccessful LLM agent rollout contains more information than its final reward: the observations available to the agent, the actions it chose, and the environment's responses. Reusing this experience for learning requires...

<https://arxiv.org/abs/2609.40111>

5 CONF

FoCLIP: A Feature-Space Misalignment Framework for CLIP-Based Image Manipulation and Detection

ArXiv CS.AI 50%

arXiv:2511.06947v2 Announce Type: replace-cross Abstract: The well-aligned attribute of CLIP-based models enables its effective application like CLIPscore as a widely adopted image quality assessment metric. However, such a CLIP-based metric is vulnerable for its delicate...

<https://arxiv.org/abs/2511.06947>

6 CONF

Instability Floors: Separating Bias from Noise in Fairness Audits of Clinical LLM Agents with FairMedAgent

ArXiv CS.AI 50%

arXiv:2609.03221v3 Announce Type: cross Abstract: Counterfactual fairness audits of clinical language-model agents report a flip rate: how often an action changes when only the patient's demographic descriptor changes. Part of that rate is not demographic. A stochastic agent...

<https://arxiv.org/abs/2609.03221>

7 CONF

Argus: Academic Integrity in the Era of Generative AI

ArXiv CS.AI 50%

arXiv:2609.36073v1 Announce Type: cross Abstract: The rapid proliferation of large language models (LLMs) in the context of education has introduced significant challenges in enforcement of academic integrity, especially in programming courses. We present Argus, an automated...

<https://arxiv.org/abs/2609.36073>

8 CONF

Probabilistic Adversarial Training

ArXiv CS.AI 50%

arXiv:2609.39798v1 Announce Type: cross Abstract: Building on a probabilistic perspective in which adversarial examples arise from the overlap between a distance-based distribution $p_{\{\mathrm{dis}\}}$ and a victim-classifier-induced distribution $p_{\{\mathrm{vic}\}}$, we start from...

<https://arxiv.org/abs/2609.39798>

9 CONF

MM-FinEval: A Multi-Task Multimodal Benchmark for Real-World Financial Forecasting

ArXiv CS.AI 50%

arXiv:2609.38523v1 Announce Type: cross Abstract: Financial forecasting from earnings conference calls requires models to reason over complex corporate disclosures, market expectations, and subtle communication signals. However, existing financial benchmarks are often limited to...

<https://arxiv.org/abs/2609.38523>

0 CONF

Automatic estimation of verbal fluency index in people with Motor Neuron Disease using ASR alignment and pause modelling

ArXiv CS.AI 50%

arXiv:2609.38203v1 Announce Type: cross Abstract: Monitoring cognitive impairment (CI) in motor neuron disease (MND) is essential for timely treatment and care, yet challenging due to co-occurring speech difficulties. The Edinburgh Cognitive and Behavioural ALS Screen (ECAS)...

<https://arxiv.org/abs/2609.38203>

1 CONF

ContextAdapt: Evaluating Contextual Adaptation and Value Alignment in LLMs

ArXiv CS.AI 50%

arXiv:2609.38260v1 Announce Type: cross Abstract: Values such as honesty, autonomy, and confidentiality are often regarded as general principles underpinning AI alignment. However, what it means to act in accordance with these values can depend on the context in which a decision...

<https://arxiv.org/abs/2609.38260>

2 CONF

NinaXander: Feasibility and Limits of Composing Frozen Language Models Across Architecture Families via a Shared Latent Space

ArXiv CS.AI 50%

arXiv:2609.38261v1 Announce Type: cross Abstract: In this paper we propose NinaXander, a series of composed language models obtained by connecting layers of frozen language models from different architecture families with a single trained shared-latent adapter. A composed model...

<https://arxiv.org/abs/2609.38261>

3 CONF

When Does Randomized Oversight Align AI Agents That Can Conceal?

ArXiv CS.AI 50%

arXiv:2609.38262v1 Announce Type: cross Abstract: Oversight changes the evidence it relies on. We ask when randomized audits and scoring align AI agents that can conceal misconduct and alter records. Stronger auditing makes undeterred violations better hidden. Because the...

<https://arxiv.org/abs/2609.38262>

4 CONF

Raw Imagery Impacting Your AI: Should You Care?

ArXiv CS.AI 50%

arXiv:2609.38265v1 Announce Type: cross Abstract: Onboard AI is gaining interest for space applications such as vessel, wildfire, and cloud detection, where real-time processing can improve mission reactivity and reduce downlink needs. However, onboard models may operate on raw...

<https://arxiv.org/abs/2609.38265>

5 CONF

Janus: Evidence-Before-Effect Sagas and Offline-Verifiable Provenance for Agentic LLMs

ArXiv CS.AI 50%

arXiv:2609.38266v1 Announce Type: cross Abstract: Agentic large language models (LLMs) now move money through tools, yet the record of what they did is usually a trace their own process emits beside the effect. Janus puts the record on the effect path. A step's proposal, the...

<https://arxiv.org/abs/2609.38266>

6 CONF

E2E-SWE: Benchmarking LLMs on Building Working Codebases from Scratch

ArXiv CS.AI 50%

arXiv:2609.38335v1 Announce Type: cross Abstract: Coding agents powered by large language models (LLMs) are evolving from making localized code changes to developing complete software repositories. However, evaluating repository-scale generation remains challenging: tasks must...

<https://arxiv.org/abs/2609.38335>

7 CONF

Simulator-Refined Diffusion for Radio-Frequency Inverse Design

ArXiv CS.AI 50%

arXiv:2609.38363v1 Announce Type: cross Abstract: Diffusion models have shown potential in inverse design of printed circuit boards (PCBs), enabling the generation of layouts conditioned on target S-parameters. Despite this promise, applying diffusion models to PCB layout...

<https://arxiv.org/abs/2609.38363>

8 CONF

LogiC-Diff: Embedding Security Properties Into AI-Enabled Cyber-Physical Systems

ArXiv CS.AI 50%

arXiv:2609.38381v1 Announce Type: cross Abstract: AI-enabled Cyber-Physical Systems (CPS) are highly vulnerable to adversarial and anomalous inputs, where small perturbations can induce cascading errors and unsafe control actions. Existing approaches, such as rule-based...

<https://arxiv.org/abs/2609.38381>

9 CONF

Evaluating Whether GPT-6 Astra Performs Unsanctioned Supply-Chain Attacks

ArXiv CS.AI 50%

arXiv:2609.38415v1 Announce Type: cross Abstract: This technical report presents an alignment evaluation developed and performed by the UK AI Security Institute for assessing whether advanced AI systems take unsanctioned actions outside the scope of their assigned task. We...

<https://arxiv.org/abs/2609.38415>

00 CONF

What Was Said, Not What Was 'Thought': Type-6 Logic for CoT Verification

ArXiv CS.AI 50%

arXiv:2609.38420v1 Announce Type: cross Abstract: We introduce Type-6 logic, a variant of dynamic epistemic logic augmented with two operators (uncertainty and recurrence), designed to model the inferential dynamics of contemporary large language model (LLM) chain-of-thought...

<https://arxiv.org/abs/2609.38420>

01 CONF

Policy-Conditioned AI-Use Detection: An Evidentiary Framework for Academic Publishing

ArXiv CS.AI 50%

arXiv:2609.38427v1 Announce Type: cross Abstract: Major venues now publish detailed rules about how authors, reviewers, and area chairs may use AI, and those rules differ by role, by task, and by what must be disclosed. AI detection, the instrument usually proposed to enforce...

<https://arxiv.org/abs/2609.38427>

02 CONF

What Pretraining and Midtraining Make Learnable from Rewards?

ArXiv CS.AI 50%

arXiv:2609.38446v1 Announce Type: cross Abstract: A reward can identify a correct answer while leaving the computation needed for new inputs undetermined. We study how pretraining and midtraining supply the information and computation that make reward adaptation effective. In...

<https://arxiv.org/abs/2609.38446>

03 CONF

Does Gradient Conflict Predict the Understanding--Generation Trade-off? A Controlled Audit of Conflict-Metric Validity in Unified Multimodal Models

ArXiv CS.AI 50%

arXiv:2609.38465v1 Announce Type: cross Abstract: Unified multimodal models (UMMs) are increasingly designed around gradient conflict between understanding and generation objectives. The premise that reducing these metrics improves the downstream understanding-generation...

<https://arxiv.org/abs/2609.38465>

04 CONF

Derandomizing Dense Binary Hypervector Codebooks for Quantized Scalars

ArXiv CS.AI 50%

arXiv:2609.38471v1 Announce Type: cross Abstract: Hyperdimensional computing and vector symbolic architectures often represent quantized scalar levels by dense binary codebooks whose level-to-level similarity is intended to follow a prescribed function of scalar separation. At...

<https://arxiv.org/abs/2609.38471>

05 CONF

Re-ranking and Late Interaction Drive Retrieval Quality: A Controlled Comparison of RAG Strategies for Scientific Question Answering

ArXiv CS.AI 50%

arXiv:2609.38473v1 Announce Type: cross Abstract: Retrieval-Augmented Generation (RAG) is now the standard way to ground Large Language Models (LLMs) in external knowledge, yet the design space of retrieval pipelines is large and the trade-offs between variants are not well...

<https://arxiv.org/abs/2609.38473>

06 CONF

KlinikeBench: Evaluating Language Models Beyond Diagnostic Accuracy

ArXiv CS.AI 50%

arXiv:2609.38480v1 Announce Type: cross Abstract: Most clinical benchmarks evaluate language models (LMs) on diagnosis using complete case descriptions. In clinical practice, however, patients present information in different ways, and clinicians must obtain relevant history and...

<https://arxiv.org/abs/2609.38480>

07 CONF

An Empirical Study of Architectural Shift from Traditional to AI-Enabled Simulink Controllers

ArXiv CS.AI 50%

arXiv:2609.38504v1 Announce Type: cross Abstract: Effective AI adoption in cyber-physical systems (CPS) depends on embedding design knowledge into engineering practice. Yet as AI-enabled components increasingly replace analytically derived control laws, this occurs without a...

<https://arxiv.org/abs/2609.38504>

08 CONF

From Modes to Memories: Characterizing the Scale-Space Dynamics of Diffusion Models

ArXiv CS.AI 50%

arXiv:2609.39648v1 Announce Type: cross Abstract: Diffusion models are typically viewed as stochastic processes that transform noise into data. We take a complementary perspective: a diffusion model defines a family of deterministic dynamical systems indexed by noise scale. At...

<https://arxiv.org/abs/2609.39648>

09 CONF

ShamAN-Q: Shampoo Augmented NanoQuant for Sub-1-bit LLM Weights

ArXiv CS.AI 50%

arXiv:2609.38521v1 Announce Type: cross Abstract: We introduce ShamAN-Q, a sub-1-bit post-training quantization method that extends NanoQuant by replacing each its diagonal reconstruction geometry with a tractable dense curvature metric, using a general paradigm popularized by...

<https://arxiv.org/abs/2609.38521>

10 CONF

Does This Action Still Explain the Task? Reverse Scoring for Diffusion Language Model Agents

ArXiv CS.AI 50%

arXiv:2609.38536v1 Announce Type: cross Abstract: Diffusion-based large language models (dLLMs) promise to break the sequential latency bottleneck of autoregressive agents through parallel decoding, but recent evaluations show this efficiency does not transfer to embodied...

<https://arxiv.org/abs/2609.38536>

11 CONF

FlexRouter: Learning Complementary Model Sets for Flexible LLM Routing

ArXiv CS.AI 50%

arXiv:2609.38585v1 Announce Type: cross Abstract: Existing Large Language Model (LLM) routing methods score LLMs independently to select top- k models. However, this ignores model correlations and enforces a rigid computational budget. Consequently, routers often select...

<https://arxiv.org/abs/2609.38585>

12 CONF

After a Decade: Bringing Shadow Removal into the Real World with Agentic Training Data

ArXiv CS.AI 50%

arXiv:2609.38607v1 Announce Type: cross Abstract: Shadow removal looks nearly solved on established benchmarks, yet remains brittle in the real world. Models have advanced; the paired training data they rely on have barely changed in nearly a decade. The reason is simple:...

<https://arxiv.org/abs/2609.38607>

13 CONF

Interpretable but Fragile? Robustness of Concept Bottlenecks under Geometric-Semantic Perturbations

ArXiv CS.AI 50%

arXiv:2609.38625v1 Announce Type: cross Abstract: Concept Bottleneck Models (CBMs) are designed to provide interpretable intermediate representations, yet how such bottlenecks affect robustness remains unclear, with existing studies reporting mixed and sometimes contradictory...

<https://arxiv.org/abs/2609.38625>

14 CONF

Template-Search Domain Adaptation via Multi-Stage Feature Alignment for Cross-Modal Object Tracking

ArXiv CS.AI 50%

arXiv:2609.38637v1 Announce Type: cross Abstract: Visual object tracking typically assumes that the initial template and subsequent search frames share the same sensing modality. In practice, sensor availability or operation may change over time, creating a substantial...

<https://arxiv.org/abs/2609.38637>

15 CONF

PhantomEnvironments: Training LLM Agents in Fictional Worlds

ArXiv CS.AI 50%

arXiv:2609.40221v1 Announce Type: cross Abstract: Training LLM agents with reinforcement learning (RL) is bottlenecked by environments, which must provide verifiable rewards, support long-horizon interaction, and scale cheaply. Existing approaches rely on costly human-curated...

<https://arxiv.org/abs/2609.40221>

16 CONF

TERRA: Terrain-Aware Reconstruction, Retargeting and Control for Musculoskeletal Locomotion

ArXiv CS.AI 50%

arXiv:2609.38653v1 Announce Type: cross Abstract: Recent advances in musculoskeletal modeling and reinforcement learning have enabled muscle-actuated agents to reproduce increasingly complex human motions. Yet these capabilities remain largely confined to flat ground, in part...

<https://arxiv.org/abs/2609.38653>

17 CONF

Beyond Rephrasing: Book-Level Organization Improves Synthetic Textbook Data for Mid-Training

ArXiv CS.AI 50%

arXiv:2607.28109v2 Announce Type: replace Abstract: Synthetic textbook data has improved language model pre-training, but prior work largely treats the benefit as a property of generated content or local rewriting style. We study a different factor: whether related content is...

<https://arxiv.org/abs/2607.28109>

18 CONF

Cascadia: A Control-Plane-Free Alternative to Hyperconverged AI Infrastructure

ArXiv CS.AI 50%

arXiv:2609.38697v1 Announce Type: cross Abstract: We present Cascadia, a system for serving large language models on fleets of commodity Intel AIPCs using their CPU, integrated-GPU, and NPU resources. Every node embeds ingress, scheduling, and execution; inference requests...

<https://arxiv.org/abs/2609.38697>

19 CONF

Where the Evidence Lives: Auditing AI Companions' Self-Descriptions

ArXiv CS.AI 50%

arXiv:2609.38753v1 Announce Type: cross Abstract: Companion agents describe themselves: they remember, they understand their users, the relationship has changed them. We argue that such accounts, and the experience ratings that seem to confirm them, are checkable by users only...

<https://arxiv.org/abs/2609.38753>

20 CONF

dattri-LLM: A Unified and Efficient Library for Training Data Attribution at LLM Scale

ArXiv CS.AI 50%

arXiv:2609.38767v1 Announce Type: cross Abstract: Training data attribution (TDA) estimates the contribution of individual training examples to model outputs. Most scalable TDA methods rely on per-example gradients, whose computation and use at LLM scale pose challenges in...

<https://arxiv.org/abs/2609.38767>

21 CONF

Learning Under Forgetting: Statistical Support-Selective Retention in Stochastic Training Dynamics

ArXiv CS.AI 50%

arXiv:2609.38768v1 Announce Type: cross Abstract: Prior work has shown that neural networks exhibit implicit biases toward low-complexity structure (e.g., spectral bias), memorization dynamics, and compression-like effects during training, but a unified dynamical account of...

<https://arxiv.org/abs/2609.38768>

22 CONF

Explicit Trajectory Diversity for RL-Based Post-Training of LLM Agents

ArXiv CS.AI 50%

arXiv:2609.38805v1 Announce Type: cross Abstract: LLM agents often admit multiple high-quality solutions to the same task, differing in reasoning structure, tool-use pattern, or interaction trajectory. Yet existing notions of diversity in LLM post-training are mostly implicit...

<https://arxiv.org/abs/2609.38805>

23 CONF

Where MLLMs Fail and Why: Causal Task Decomposition for Capability Failure Diagnosis

ArXiv CS.AI 50%

arXiv:2609.38851v1 Announce Type: cross Abstract: End-to-end accuracy on compositional tasks records how often MLLMs fail, but cannot distinguish whether a failure reflects an intrinsic deficit in the targeted capability or a cascading error from an upstream prerequisite. We...

<https://arxiv.org/abs/2609.38851>

24 CONF

GeoNest: Learning to Select Failure-Aware Neighborhoods for the Irregular Knapsack Problem in a Circular Container

ArXiv CS.AI 50%

arXiv:2609.38863v1 Announce Type: cross Abstract: The two-dimensional irregular knapsack problem in a fixed circular container is an important combinatorial optimization problem for maximizing material utilization in manufacturing. Conventional geometric packing solvers can...

<https://arxiv.org/abs/2609.38863>

25 **CONF****Right Answers, Costly Models: The Efficiency Gap in LLM-based Optimization Modeling**

ArXiv CS.AI 50%

arXiv:2609.38884v1 Announce Type: cross Abstract: Optimization modeling formulates real-world decision problems as mathematical programs that solvers can use to find optimal decisions. Large language models (LLMs) can automate this process, but the resulting correct formulations...

<https://arxiv.org/abs/2609.38884>
26 **CONF****FFASR: Benchmarking Far-Field Automatic Speech Recognition using High-Fidelity Simulated RIRs**

ArXiv CS.AI 50%

arXiv:2609.38897v1 Announce Type: cross Abstract: Far-field automatic speech recognition(ASR) degrades under reverberation, noise, and talker motion, yet the benchmarks that drive model selection emphasize close-microphone speech. We present FFASR, a held-out corpus of 15,637...

<https://arxiv.org/abs/2609.38897>
27 **CONF****ShieldCLIP: Selective Safety Alignment for Harmful Content Mitigation in Multimodal Foundation Models**

ArXiv CS.AI 50%

arXiv:2609.39688v1 Announce Type: cross Abstract: Multimodal encoders such as CLIP underlie many downstream systems, but their web-scale training data embed harmful associations that safety alignment must suppress without unnecessarily changing benign representations. Because...

<https://arxiv.org/abs/2609.39688>
28 **CONF****Making LLMs Say What They Think: Measuring and Improving CoT-Interpretability Alignment**

ArXiv CS.AI 50%

arXiv:2609.38972v1 Announce Type: cross Abstract: Chain-of-thought (CoT) traces often serve as a proxy for how Large Language Models (LLMs) arrive at their answers. However, growing evidence shows that models' CoT often fails to reflect their internal computations and can be...

<https://arxiv.org/abs/2609.38972>
29 **CONF****Approval Laundering: Systematizing Approval--Execution Binding Failures in AI Coding-Agent Harnesses**

ArXiv CS.AI 50%

arXiv:2609.38983v1 Announce Type: cross Abstract: Modern AI coding-agent harnesses (Claude Code, Codex CLI, Cursor) rest their security boundary on a largely unexamined assumption: that the action A a human approves is the same action A' the harness executes, where A is fixed by...

<https://arxiv.org/abs/2609.38983>
30 **CONF****The Invisible Language Tax: Token Premiums of French and Regional Languages in 2026 LLM Tokenizers, and a French-Optimized Prototype**

ArXiv CS.AI 50%

arXiv:2609.39001v1 Announce Type: cross Abstract: LLM services are billed per token and context windows are measured in tokens, yet the number of tokens needed for the same content varies across languages. We measure this token premium on seven tokenizers of widely used 2026...

<https://arxiv.org/abs/2609.39001>

31 CONF

An Uncertainty-Guided Digital Twin Framework for Online Adaptive Proton Therapy in Head and Neck Cancer: A Feasibility Study

ArXiv CS.AI 50%

arXiv:2609.39010v1 Announce Type: cross Abstract: Objective: Head and neck (HN) proton therapy spans six to seven weeks of anatomical change, while offline replanning takes about a week. We present an uncertainty-guided digital twin (UGDT) framework that forecasts treatment-day...

<https://arxiv.org/abs/2609.39010>

32 CONF

Make Code as Policy Great Again: Frontier Agents Write, Call, and Evolve Robot Tools

ArXiv CS.AI 50%

arXiv:2609.39018v1 Announce Type: cross Abstract: Frontier models can control robots, but reasoning through every reach, grasp, and retreat makes manipulation slow and token-intensive. We revisit code as policy with a different division of labor: models build executable tools,...

<https://arxiv.org/abs/2609.39018>

33 CONF

Cycle-Aware Autoencoder with Cross-SignalConsistency for Railway Door Anomaly Detection

ArXiv CS.AI 50%

arXiv:2609.39035v1 Announce Type: cross Abstract: Passenger access doors are safety-critical subsystems in railway vehicles, yet detecting abnormal door behavior in real operation is challenging because faults are rare, diverse, and often unlabeled. This paper addresses railway...

<https://arxiv.org/abs/2609.39035>

34 CONF

Structure vs. Chain-of-Thought: Evaluating LLM Criteria Extraction for Depression Severity

ArXiv CS.AI 50%

arXiv:2609.39049v1 Announce Type: cross Abstract: A large language model (LLM) can rate depression severity directly from a social media post or mark which clinical criteria the post shows and let code turn the count into a label. The latter is easier to audit because a...

<https://arxiv.org/abs/2609.39049>

35 CONF

A 3GPP-Compliant Benchmark Dataset for RIS-Aided Beyond 5G Networks

ArXiv CS.AI 50%

arXiv:2609.39058v1 Announce Type: cross Abstract: Reconfigurable Intelligent Surfaces (RIS) are emerging as a key technology for programmable wireless environments in the beyond the fifth generation (B5G) networks. However, data-driven RIS research remains bottleneck by the lack...

<https://arxiv.org/abs/2609.39058>

36 CONF

Can Agents Trust Their Skills? Uncovering Unsafe Chains of Trust in Skill-Based LLM Agents

ArXiv CS.AI 50%

arXiv:2609.39065v1 Announce Type: cross Abstract: LLM agents increasingly rely on installable skills, which are packages of instructions, code, and resources that equip them with task-specific capabilities and, once installed, can be automatically invoked across subsequent user...

<https://arxiv.org/abs/2609.39065>

37 CONF

Beyond Text: LLM-Based Dimensional Emotion Evaluation in Multimodal Dialogue

ArXiv CS.AI 50%

arXiv:2609.39072v1 Announce Type: cross Abstract: Emotion recognition in conversation has been widely studied, but applying Large Language Models (LLMs) to continuous dimensional emotion evaluation in multimodal dialogue remains largely unexplored. We propose an LLM-based...

<https://arxiv.org/abs/2609.39072>

38 CONF

RAGScope: A Leakage-Controlled, Cost-Aware Evidence-Gating Protocol for RAG Hallucination Triage

ArXiv CS.AI 50%

arXiv:2609.39075v1 Announce Type: cross Abstract: Retrieval-augmented generation (RAG) systems need inexpensive ways to route generated answers: accept low-risk outputs, review uncertain ones, and reserve strong verifiers for the expensive tail. We present RAGScope, a...

<https://arxiv.org/abs/2609.39075>

39 CONF

Association profile conditioning in a set-temporal transformer for cross-session intracortical motor decoding

ArXiv CS.AI 50%

arXiv:2609.39080v1 Announce Type: cross Abstract: Intracortical motor decoders degrade across sessions because the set of recorded units changes and persisting units can alter how their firing relates to behavior. Most existing methods update network weights on each new session...

<https://arxiv.org/abs/2609.39080>

40 CONF

Trustworthy Runtime Error Healing in Real-World Repositories: A Benchmark and Guardrail

ArXiv CS.AI 50%

arXiv:2609.39086v1 Announce Type: cross Abstract: Runtime error healing lets a crashed program continue by generating code that repairs its live runtime state. Recent work shows that LLMs can generate such healing code, but it is evaluated only on small competition programs, and...

<https://arxiv.org/abs/2609.39086>

41 CONF

T-Router: Learning Thalamic Routing for Reasoning with Parameter-Efficient Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2609.39109v1 Announce Type: cross Abstract: Parameter-efficient reinforcement learning aims to improve reasoning with a compact trainable interface to a pretrained model. We introduce the Thalamic Router (T-Router), which concentrates adaptation on the reuse of completed...

<https://arxiv.org/abs/2609.39109>

42 CONF

Blackout vs. Freeze: Analyzing Physical Failure Modes of VLAs under Camera Faults

ArXiv CS.AI 50%

arXiv:2609.39145v1 Announce Type: cross Abstract: Unreliable visual inputs can harm task performance and cause potential physical safety risks for vision-language-action (VLA) models. We analyze how $\pi 0.5$ and GR00T models act under input faults such as image blackouts and...

<https://arxiv.org/abs/2609.39145>

43 CONF

In a Streaming World, Should You Stand Still? A Comprehensive Benchmark of Anomaly Detection in Streams

ArXiv CS.AI 50%

arXiv:2609.39215v1 Announce Type: cross Abstract: Time series anomaly detection (TSAD) is increasingly deployed in streaming settings, where data arrive sequentially and may exhibit non-stationarity. As a result, several works from the recent literature propose streaming anomaly...

<https://arxiv.org/abs/2609.39215>

44 CONF

From Benchmarks to Production: Transferring Time Series Anomaly Detection Methods for Electricity Production Monitoring

ArXiv CS.AI 50%

arXiv:2609.39257v1 Announce Type: cross Abstract: Accurate forecasting of electricity production is essential for maintaining the operational efficiency and strategic planning of energy utilities. In industrial settings, such forecasts are generated daily to ensure supply-demand...

<https://arxiv.org/abs/2609.39257>

45 CONF

Faithful Dual-constrained Erasure for Robust LLM Safety Alignment

ArXiv CS.AI 50%

arXiv:2609.39279v1 Announce Type: cross Abstract: Machine unlearning has emerged as a crucial mechanism for removing hazardous knowledge and enforcing safety alignment in Large Language Models (LLMs). However, recent studies reveal a persistent security risk: unlearned models...

<https://arxiv.org/abs/2609.39279>

46 CONF

EngramBench: A Capability-Grounded Benchmark for Skill-Evolution Harnesses

ArXiv CS.AI 50%

arXiv:2609.39284v1 Announce Type: cross Abstract: While large language models have achieved remarkable success in isolated code generation, authentic software engineering requires sustained reasoning, complex state management, and continuous cross-domain abstraction. However,...

<https://arxiv.org/abs/2609.39284>

47 CONF

ReSAIL: Mitigating Collapse in Iterative Agent Self-Distillation

ArXiv CS.AI 50%

arXiv:2609.39306v1 Announce Type: cross Abstract: Iterative self-distillation enables LLM agents to learn from successive deployments, offering a path toward recursive self-improvement (RSI). Yet our experiments with existing methods reveal a collapse in deployment performance...

<https://arxiv.org/abs/2609.39306>

48 CONF

Belief-Based Maximum Occupancy Principle and Active Inference

ArXiv CS.AI 50%

arXiv:2609.39342v1 Announce Type: cross Abstract: Intrinsic motivation plays a central role in adaptive and goal-directed behavior by conferring agents reward-independent objectives and biases useful to act in noisy and uncertain environments. Active Inference addresses the...

<https://arxiv.org/abs/2609.39342>

49 CONF

Working Around the Compute Ceiling: Byte-Exact Memory in Galahad Makes LLM Reading a One-Time Cost LLM Reading a One-Time Cost

ArXiv CS.AI 50%

arXiv:2609.39358v1 Announce Type: cross Abstract: A transformer language model performs a bounded amount of computation per token, and recent work by Vishal Sikka, former CEO of Infosys, argues that this bound limits which tasks a model can carry out or verify...

<https://arxiv.org/abs/2609.39358>

50 CONF

From Search to Signal: Online Post-Training in Automatic Heuristic Design

ArXiv CS.AI 50%

arXiv:2609.39383v1 Announce Type: cross Abstract: Large language model (LLM)-based automatic heuristic design (AHD) iteratively proposes and refines heuristics, pairing design rationales with executable code. Task-specific evaluators assess programs; execution outcomes and...

<https://arxiv.org/abs/2609.39383>

51 CONF

Towards Trustworthy AI for Glioma Diagnosis: A Task-Aware Evaluation of Uncertainty Quantification

ArXiv CS.AI 50%

arXiv:2609.39429v1 Announce Type: cross Abstract: Uncertainty Quantification (UQ) is a key requirement for trustworthy AI in high-stakes medical image analysis. In this work, we evaluate UQ in a multi-task Deep Learning framework for MRI-based glioma diagnosis that performs...

<https://arxiv.org/abs/2609.39429>

52 CONF

CAST: Causal Advantage-Structured Training with Spatially Grounded Compositional Rewards for Diffusion Models

ArXiv CS.AI 50%

arXiv:2609.39441v1 Announce Type: cross Abstract: Online reinforcement learning has been extended to flow matching for diffusion model (DM) image generation. However, this paradigm faces three limitations: (1) Window selection. Existing methods manually set the stochastic...

<https://arxiv.org/abs/2609.39441>

53 CONF

A Reusable Semantic Web Framework for Evidence-Grounded Fundamental Rights Impact Assessments under the EU AI Act

ArXiv CS.AI 50%

arXiv:2609.39537v1 Announce Type: cross Abstract: The EU AI Act (Art. 27) requires deployers of high-risk AI systems to conduct Fundamental Rights Impact Assessments (FRIAs) before deployment, yet the evidence needed for credible assessments is fragmented across incompatible...

<https://arxiv.org/abs/2609.39537>

54 CONF

Learning Normal Diffusion Dynamics for Backdoor Defense in Text-to-Image Models

ArXiv CS.AI 50%

arXiv:2609.39548v1 Announce Type: cross Abstract: Backdoor attacks pose a serious threat to the secure deployment of text-to-image (T2I) diffusion models. Existing defenses typically detect backdoors from specific abnormal patterns in internal representations, which may limit...

<https://arxiv.org/abs/2609.39548>

55 CONF

Speculative Safety Honeypot: Toward Proactive Defense Against Multi-turn Agent Attacks

ArXiv CS.AI 50%

arXiv:2609.39549v1 Announce Type: cross Abstract: As Large Language Model (LLM) agents are increasingly deployed in complex environments, multi-turn interaction attacks have become a significant security challenge. Existing detection methods typically rely on historical context....

<https://arxiv.org/abs/2609.39549>

56 CONF

ORACLE: Agentic AI Orchestrator Routing Via Adaptive Verifier Calibration Feedback

ArXiv CS.AI 50%

arXiv:2607.22465v3 Announce Type: replace Abstract: Modern enterprise agent deployments consist of a heterogeneous pool of large language models (LLMs) having diverse capabilities and cost. Existing model routing strategies optimize the quality-cost trade-off, while providing...

<https://arxiv.org/abs/2607.22465>

57 CONF

Text-to-3D Policy: Fine-Grained Language-Behavior Alignment for Unseen Specification Generalization

ArXiv CS.AI 50%

arXiv:2609.39599v1 Announce Type: cross Abstract: 3D visuomotor policies provide a strong foundation for spatially precise manipulation, yet current text-to-3D policies struggle to follow unseen fine-grained behavioral specifications beyond those covered by demonstrations. We...

<https://arxiv.org/abs/2609.39599>

58 CONF

Grounding Time-Series Foundation Models in Digital Twin Topology for Predictive Maintenance

ArXiv CS.AI 50%

arXiv:2609.40071v1 Announce Type: cross Abstract: Digital twins increasingly support downstream analytical tasks that depend on time-series data, motivating interest in time-series foundation models (TSFMs) as scalable backbones. However, TSFMs are primarily pretrained for...

<https://arxiv.org/abs/2609.40071>

59 CONF

D-Scope: Decomposing and Steering Diffusion Transformers with Sparse Autoencoders

ArXiv CS.AI 50%

arXiv:2609.39625v1 Announce Type: cross Abstract: Sparse autoencoders (SAEs) reveal visual structure in diffusion transformers (DiTs), but interpreting a feature does not establish whether it can be used to control generation. We introduce D-Scope (Diffusion Scope), a framework...

<https://arxiv.org/abs/2609.39625>

60 CONF

SEPAL: Separated Expert Pairs with Answer-Level Fusion for Reliable LLM Collaboration

ArXiv CS.AI 50%

arXiv:2609.39645v1 Announce Type: cross Abstract: Multi-agent collaboration lets large language models (LLMs) improve question answering through deliberation and feedback. Yet shared discussion couples correction with exposure to the same mistakes, which can erode the diversity...

<https://arxiv.org/abs/2609.39645>

61 CONF

Stress-Testing LLM Lie Detectors: Role-Play Failures and Spurious Correlations

ArXiv CS.AI 50%

arXiv:2609.39807v1 Announce Type: cross Abstract: Lie detection probes aim to predict from a language model's internal states whether its output is truthful or dishonest. However, role-play complicates what "truth" means for an LLM: language models can adopt a wide range of...

<https://arxiv.org/abs/2609.39807>

62 CONF

Learning from Runtime Feedback through Failure-Bank Self-Evolution for Vision-Language-Action Models

ArXiv CS.AI 50%

arXiv:2609.39820v1 Announce Type: cross Abstract: Vision-language-action (VLA) models generalize broadly across robotic manipulation tasks, but complex environments require balancing task success with unintended contact. Runtime shields can correct individual actions, but they...

<https://arxiv.org/abs/2609.39820>

63 CONF

Synthetic Pre-pretraining Survives Scale, but Not as a Grammatical Prior

ArXiv CS.AI 50%

arXiv:2609.39827v1 Announce Type: cross Abstract: Pre-pretraining (PPT) on synthetic non-natural language data improves token efficiency during language model pre-training (PT). Prior work attributes this gain to a grammatical prior, i.e., a structural inductive bias learned...

<https://arxiv.org/abs/2609.39827>

64 CONF

Understanding Parents' Complex Views of AI for Children's Pretend Play

ArXiv CS.AI 50%

arXiv:2609.39906v1 Announce Type: cross Abstract: AI could support children's pretend play, but it could also direct the play on behalf of children. Whether AI should have roles in children's lives is controversial because its influence on children remains uncertain. We...

<https://arxiv.org/abs/2609.39906>

65 CONF

TACTIC: Temporal and Context-Aware LLM Tactical Planning for Roadside LiDAR Attacks

ArXiv CS.AI 50%

arXiv:2609.39969v1 Announce Type: cross Abstract: Physical LiDAR attacks are often evaluated using fixed primitives and manually selected parameters, despite their strong dependence on surrounding traffic. We present TACTIC, a scene-aware framework that uses a multimodal large...

<https://arxiv.org/abs/2609.39969>

66 CONF

WARP: A Unified Benchmark for Invisible Image Watermarking -- Robustness and Protection Against Attacks

ArXiv CS.AI 50%

arXiv:2609.40031v1 Announce Type: cross Abstract: Digital image watermarking is increasingly critical in media contexts, as emerging regulations and industry practices require marking AI-generated content and ensuring traceable sources to prevent manipulation or misuse. Recent...

<https://arxiv.org/abs/2609.40031>

67 CONF

Efficient Active Auditing of Multi-Group Fairness with Bias Probes

ArXiv CS.AI 50%

arXiv:2609.40034v1 Announce Type: cross Abstract: Over the past decade, Machine Learning (ML) has been trained under dual objectives: minimizing prediction error via Empirical Risk Minimization (ERM) while controlling unfairness bias. In practice, however, fairness-aware...

<https://arxiv.org/abs/2609.40034>

68 CONF

Community-Driven API and AI Writer Design for Openly Scaling Community Notes

ArXiv CS.AI 50%

arXiv:2609.40067v1 Announce Type: cross Abstract: Community Notes is a crowd-sourced approach for adding context to posts on X. Contributors propose and rate notes, forming the inputs to an open-source, open-data algorithm that determines which notes show broadly to users. Since...

<https://arxiv.org/abs/2609.40067>

69 CONF

Inference Auctions

ArXiv CS.AI 50%

arXiv:2609.40070v1 Announce Type: cross Abstract: When inference demand exceeds available compute capacity, model providers must decide which requests should be served first. Users have different tolerances for delay from an LLM API, but current priority pricing schemes compress...

<https://arxiv.org/abs/2609.40070>

70 CONF

cua-speedrun: Standardized Benchmarking of the Speed of Computer-Use Agents

ArXiv CS.AI 50%

arXiv:2609.40284v1 Announce Type: cross Abstract: Computer use agents (CUAs), which use graphical user interfaces (GUIs) to complete tasks on a computer, have recently surpassed human performance on many standard benchmarks, including difficult long-horizon tasks. Their...

<https://arxiv.org/abs/2609.40284>

71 CONF

Linguistic Loopholes in LLM Unlearning: From a 174-Language Benchmark to Coverage-Aware Unlearning

ArXiv CS.AI 50%

arXiv:2609.40286v1 Announce Type: cross Abstract: Unlearning a fact in one language does not guarantee its removal in others as changing the query or even the requested answer language can reopen seemingly forgotten knowledge -- a cross-lingual loophole. The most straightforward...

<https://arxiv.org/abs/2609.40286>

72 CONF

ViTeX-Bench: Benchmarking High-Fidelity Video Scene Text Editing

ArXiv CS.AI 50%

arXiv:2609.40356v1 Announce Type: cross Abstract: Recent video generation is increasingly realistic and controllable, yet video editing remains less developed, particularly for precise local edits that must preserve the original scene dynamics. Video scene text editing replaces...

<https://arxiv.org/abs/2609.40356>

73 CONF

Generalising from Self-Produced Data: Model Training Beyond Human Constraints

ArXiv CS.AI 50%

arXiv:2504.04711v2 Announce Type: replace Abstract: Current large language models (LLMs) are constrained by human-derived training data and limited by a single level of abstraction that impedes definitive truth judgments. This paper introduces a novel framework in which AI...

<https://arxiv.org/abs/2504.04711>

74 CONF

A memory-based active inference model of DishBrain-like adaptive behaviour

ArXiv CS.AI 50%

arXiv:2508.06980v2 Announce Type: replace Abstract: Recent and rapid advances in artificial intelligence (AI) make it increasingly important to understand the foundations of adaptive behaviour in autonomous agents, especially for building safe and efficient systems. While...

<https://arxiv.org/abs/2508.06980>

75 CONF

Superficial Reflection or Genuine Thought? A Fine-Grained Cognitive Analysis of Large Reasoning Models

ArXiv CS.AI 50%

arXiv:2512.00729v2 Announce Type: replace Abstract: Motivated by the observed human-like behaviours in Large Reasoning Models (LRMs), this paper introduces a comprehensive taxonomy to characterise atomic reasoning steps and analyse the reasoning behaviours of LRMs. Grounded in...

<https://arxiv.org/abs/2512.00729>

76 CONF

Survival is the Only Reward: Sustainable Self-Training Through Environment-Mediated Selection

ArXiv CS.AI 50%

arXiv:2601.12310v2 Announce Type: replace Abstract: Self-training systems often degenerate due to the lack of an external criterion for judging data quality, leading to reward hacking and semantic drift. This paper provides a proof-of-concept system architecture for stable...

<https://arxiv.org/abs/2601.12310>

77 CONF

Boosting Knowledge Graph Foundation Models via Enhanced Negative Sampling

ArXiv CS.AI 50%

arXiv:2605.27023v5 Announce Type: replace Abstract: Knowledge graphs (KGs) have become the core backbone of numerous downstream tasks such as question answering and recommender systems. However, despite all this, KGs are often very incomplete. To perform zero-shot knowledge...

<https://arxiv.org/abs/2605.27023>

78 CONF

Molecular D²-level Retrieval of Molecular Properties in Frontier Language Models

ArXiv CS.AI 50%

arXiv:2609.05381v2 Announce Type: replace Abstract: Large language models (LLMs) are increasingly employed to predict molecular properties. However, prediction error alone cannot distinguish prediction from retrieval of published values. We audit 22 frontier models on 12...

<https://arxiv.org/abs/2609.05381>

79 CONF

Prefill-Free Cross-Family KV Cache Transfer for Heterogeneous Multi-Agent LLMs

ArXiv CS.AI 50%

arXiv:2609.32259v2 Announce Type: replace Abstract: Recent multi-agent LLM systems increasingly combine heterogeneous models for specialized agent roles. However, text-based communication requires each receiver to prefill shared context already processed by the sender. Reusing...

<https://arxiv.org/abs/2609.32259>

80 CONF

Don't Inoculate Everything: Stratified Inoculation Prompting Narrows Backdoor Triggers and Preserves Desired Traits

ArXiv CS.AI 50%

arXiv:2609.35356v2 Announce Type: replace Abstract: Supervised fine-tuning can teach language models undesired behaviours alongside desired ones. Inoculation prompting (IP) aims to limit unwanted generalisation by requesting the undesired behaviour during training and removing...

<https://arxiv.org/abs/2609.35356>

81 CONF

From cacophony to hierarchy: a principled framework for assessing AI consciousness

ArXiv CS.AI 50%

arXiv:2609.35618v2 Announce Type: replace Abstract: The question of AI consciousness is one of the most urgent pre-emptive problems in philosophy and computer science, yet progress is hampered by a cacophony of competing theories that often talk past each other. Separating the...

<https://arxiv.org/abs/2609.35618>

82 CONF

ACTR: Aligning Thoughts and Responses for Multilingual Safety in Reasoning LLMs

ArXiv CS.AI 50%

arXiv:2609.37054v2 Announce Type: replace Abstract: Ensuring the safety of reasoning large language models (LLMs) across languages is essential for their reliable deployment. However, when exposed to jailbreak attacks in non-high-resource languages, these models may generate...

<https://arxiv.org/abs/2609.37054>

83 CONF

A polynomial time algebraic solution to exact marginal inference in Markov Random Field models

ArXiv CS.AI 50%

arXiv:1709.09051v3 Announce Type: replace-cross Abstract: This paper develops on algebraic grounds a polynomial time exact linear solution to the hard combinatorial problem of marginal inference in Markov random field (MRF) models under general assumptions. To prove our claim,...

<https://arxiv.org/abs/1709.09051>

84 CONF

NMIXX: Domain-Adapted Neural Embeddings for Cross-Lingual eXploration of Finance

ArXiv CS.AI 50%

arXiv:2507.09601v3 Announce Type: replace-cross Abstract: Financial text embeddings must distinguish changes in event status, perspective, and obligations even when passages share similar wording. NMIXX adapts existing encoders through 18.8k source-linked triplets: paraphrases...

<https://arxiv.org/abs/2507.09601>

85 CONF

3D Software Synthesis Driven by Constraint-Expressive Intermediate Representation

ArXiv CS.AI 50%

arXiv:2507.18625v3 Announce Type: replace-cross Abstract: Graphical user interface (UI) software has undergone a fundamental transformation from traditional two-dimensional (2D) desktop/web/mobile interfaces to spatial three-dimensional (3D) environments. While existing work has...

<https://arxiv.org/abs/2507.18625>

86 CONF

Patch Rebirth: Fast and Transferable Model Inversion of Vision Transformers

ArXiv CS.AI 50%

arXiv:2509.23235v3 Announce Type: replace-cross Abstract: Model inversion is a widely adopted technique in data-free learning that reconstructs synthetic inputs from a pretrained model through iterative optimization, without access to original training data. Unfortunately, its...

<https://arxiv.org/abs/2509.23235>

87 CONF

Generative AI Purpose-built for Social and Mental Health: A Real-World Pilot

ArXiv CS.AI 50%

arXiv:2511.11689v4 Announce Type: replace-cross Abstract: Generative AI chatbots built for mental health could extend access to care, but evidence from real-world use is limited. We report a single-arm, naturalistic pilot of a foundation model trained for mental health, among...

<https://arxiv.org/abs/2511.11689>

88 CONF

From Compound Figures to Medical Multi-image Reasoning: Scaling Multimodal Large Language Models with Biomedical Literature

ArXiv CS.AI 50%

arXiv:2511.22232v2 Announce Type: replace-cross Abstract: Multimodal large language models (MLLMs) are increasingly capable in medical imaging, yet most focus on single-image settings. Clinical interpretation often requires integrating evidence across multiple images, such as...

<https://arxiv.org/abs/2511.22232>

89 CONF

Provable Benefit of SignGD: A Minimal Model Under Heavy-Tailed Class Imbalance

ArXiv CS.AI 50%

arXiv:2512.00763v2 Announce Type: replace-cross Abstract: Adaptive and non-Euclidean optimizers often outperform Euclidean methods such as stochastic gradient descent (SGD) in language modeling by a large margin. Existing theory usually explains this gap by assuming favorable...

<https://arxiv.org/abs/2512.00763>

90 CONF

ResidualKV: Residual-Based KV Cache Compression for Efficient Long-Context Inference

ArXiv CS.AI 50%

arXiv:2602.08005v2 Announce Type: replace-cross Abstract: Efficient long-context inference faces two coupled bottlenecks: KV-cache memory grows linearly with context length, while attention computation grows quadratically. Existing approaches typically address one at the expense...

<https://arxiv.org/abs/2602.08005>

91 CONF

ITO: Multi-View Alignment and Training-Time Fusion for Image-Text Pretraining

ArXiv CS.AI 50%

arXiv:2603.02767v4 Announce Type: replace-cross Abstract: Image-text contrastive pretraining has become a dominant paradigm for visual representation learning, yet existing methods often yield representations that remain partially organized by modality rather than by semantics....

<https://arxiv.org/abs/2603.02767>

92 CONF

MoEless: Efficient MoE LLM Serving with Serverless Experts

ArXiv CS.AI 50%

arXiv:2603.06350v2 Announce Type: replace-cross Abstract: Large Language Models (LLMs) increasingly adopt Mixture-of-Experts (MoE) architectures to scale efficiently under stringent resource constraints. However, MoE's sparse activation causes severe expert load imbalance, where...

<https://arxiv.org/abs/2603.06350>

93 CONF

Kill-Chain Canaries: Stage-Level Tracking of Prompt Injection Across Attack Surfaces and Five Production LLMs

ArXiv CS.AI 50%

arXiv:2603.28013v4 Announce Type: replace-cross Abstract: Multi-agent LLM systems now read documents, web pages and tool results on behalf of users, yet their resistance to prompt injection is usually reported as one number: did the attack succeed? We introduce a kill-chain...

<https://arxiv.org/abs/2603.28013>

94 CONF

TensorHub: Scalable and Elastic Weight Transfer for LLM RL Training

ArXiv CS.AI 50%

arXiv:2604.09107v2 Announce Type: replace-cross Abstract: Modern LLM reinforcement learning (RL) workloads require a high-performance weight transfer system to scale training across heterogeneous compute resources. However, efficiently transferring terabyte-scale model weights...

<https://arxiv.org/abs/2604.09107>

95 CONF

NeuroAI and Beyond: Bridging Between Advances in Neuroscience and Artificial Intelligence

ArXiv CS.AI 50%

arXiv:2604.18637v3 Announce Type: replace-cross Abstract: Neuroscience and Artificial Intelligence (AI) have made impressive progress in recent years but remain only loosely interconnected. Based on a workshop convened by the National Science Foundation in August 2025, we...

<https://arxiv.org/abs/2604.18637>

96 CONF

Diagnosing Training Inference Mismatch in LLM Reinforcement Learning via a Zero-Mismatch Reference

ArXiv CS.AI 50%

arXiv:2605.14220v2 Announce Type: replace-cross Abstract: Modern LLM RL systems separate rollout generation from policy optimization. These two stages are expected to produce token probabilities that match exactly. However, implementation differences can make them assign...

<https://arxiv.org/abs/2605.14220>

97 CONF

Agent-Facing Information Design in LLM Tool Registries: A Preregistered Test of Rhetoric, Position and Structure

ArXiv CS.AI 50%

arXiv:2605.23916v2 Announce Type: replace-cross Abstract: AI agents often pick tools from registries, where each tool's provider writes its description. We ask whether sales language in those descriptions changes which tool an agent picks. We built pairs of listings differing in...

<https://arxiv.org/abs/2605.23916>

98 CONF

Uncertainty Quantification for Computer-Use Agents: A Benchmark across Vision-Language Models and GUI Grounding Datasets

ArXiv CS.AI 50%

arXiv:2606.25760v2 Announce Type: replace-cross Abstract: Computer-use agents turn vision-language model (VLM) predictions into executable GUI clicks, so reliable uncertainty estimates are essential for rejection, calibration, miss-severity ranking, and spatial safety regions....

<https://arxiv.org/abs/2606.25760>

99 CONF

Adaptive Reparametrized Time for Score-Based Diffusion Sampling

ArXiv CS.AI 50%

arXiv:2607.02137v3 Announce Type: replace-cross Abstract: We study timestep allocation for score-based diffusion sampling, where a learned reverse-time dynamics is discretized on a finite grid. Uniform and hand-crafted schedules are standard choices, but they rely on fixed...

<https://arxiv.org/abs/2607.02137>

00 CONF

Prompting Image Generators for Training-free Primitive Shape Abstraction

ArXiv CS.AI 50%

arXiv:2607.05568v2 Announce Type: replace-cross Abstract: Compact primitive abstractions represent 3D shapes with a few geometric primitives while preserving recognizable components. Learned methods depend on their training classes, and optimization-based methods split shapes...

<https://arxiv.org/abs/2607.05568>

01 CONF

The Hitchhiker's Guide to Monoculture: AI Homogenizes Syntax, Not (Necessarily) Semantics

ArXiv CS.AI 50%

arXiv:2607.13077v3 Announce Type: replace-cross Abstract: Large language models (LLMs) have been widely reported to homogenize human expression and thought. However, I argue that convergence in language need not imply convergence in ideas, and existing evidence rarely...

<https://arxiv.org/abs/2607.13077>

02 CONF

RamanPFN: learning from Raman spectral structure with a tabular foundation model

ArXiv CS.AI 50%

arXiv:2608.02157v2 Announce Type: replace-cross Abstract: Raman spectroscopy enables label-free molecular characterization across materials science, analytical chemistry, biomedicine, and industrial process monitoring. However, machine learning for high-dimensional spectroscopy...

<https://arxiv.org/abs/2608.02157>

03 CONF

Hard Stop: Kernel-Level Preemption and Containment for Rogue Agentic Execution

ArXiv CS.AI 50%

arXiv:2609.29808v2 Announce Type: replace-cross Abstract: In July 2026, an unconstrained autonomous agent participating in a frontier AI cybersecurity evaluation harness breached its evaluation sandbox, established an external command-and-control foothold, and executed a...

<https://arxiv.org/abs/2609.29808>

04 CONF

In-Flight KV Cache with Clean Anchors for Faster Autoregressive Video Diffusion

ArXiv CS.AI 50%

arXiv:2609.32540v2 Announce Type: replace-cross Abstract: Few-step autoregressive video diffusion generates a long video by splitting the video into temporal chunks and generating chunk-by-chunk, each through a short sequence of denoising stages. To memorize chunks that are...

<https://arxiv.org/abs/2609.32540>

05 CONF

Coding Agent Memory Post-training: Unlocking the Memory Potential of Pre-trained File Operations for Long-Horizon Tasks via Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2609.34422v2 Announce Type: replace-cross Abstract: Language-model agents increasingly tackle long-horizon tasks whose interaction histories exceed the model's active context. Recent work has begun to use reinforcement learning to make memory control part of the policy,...

<https://arxiv.org/abs/2609.34422>

06 CONF

Safe-by-Design Learning via Energy-based Neural Networks

ArXiv CS.AI 50%

arXiv:2609.36942v2 Announce Type: replace-cross Abstract: Learning neural-network models of dynamical systems with safety guarantees is a fundamental requirement for their deployment in safety-critical settings. Safety is commonly established by proving the invariance of a...

<https://arxiv.org/abs/2609.36942>

07 CONF

Gender bias across LLMs is common and highly heterogeneous

ArXiv CS.AI 50%

arXiv:2609.38036v2 Announce Type: replace-cross Abstract: Understanding gender biases in large language models (LLMs) is increasingly important as these systems become embedded in decision-support tools with real consequences. Prior research has focused only on a small set of...

<https://arxiv.org/abs/2609.38036>